

Lesson Text

Unit I: Foundations

Module 1: What This Data Is (and Isn't)

What This Data Is (and Isn't): Situating the Minnesota Felony Sentencing Dataset

Every analytic task that follows be it filtering, summarizing, comparing groups, presupposes an answer to a fundamental question: what is the dataset students will be exploring, and what can it honestly tell them? That question is not a formality. Numbers about crime are among the most misread statistics in public life. In 23 of 27 national surveys conducted since 1993, a majority of Americans reported believing that crime was rising, even across long stretches in which recorded crime was falling (Gramlich, 2024). A statistic is only as trustworthy as one's understanding of where it came from and what it leaves out (Best, 2001). This chapter therefore traces the Minnesota felony sentencing dataset back to its origin—who created it, why, and at what point in the life of a crime it was recorded—so that a reader can state, in plain terms, which questions the roughly 294,000 felony sentencing records covering 2001 through 2019 can answer and, just as importantly, which they cannot.

Administrative Data: Built to Run a System, Not to Answer a Question

The dataset is administrative data: information generated as a by-product of running an organization rather than collected to answer a research question (Connelly et al., 2016). Courts and the Minnesota Sentencing Guidelines Commission record sentences because the state must document and monitor how people are punished, not because a researcher designed a study. That

origin is at once a strength and a limitation. Administrative records cover nearly every case, span decades, and are not subject to survey sampling error. Yet because the data was shaped by the needs of the system that produced it, using it for research requires reconstructing how each field was defined and recorded; the categories, rules, and exceptions were chosen by administrators, not by the analyst (Connelly et al., 2016). A useful discipline, then, is to treat every column as an administrative decision first and a measurement second, and to ask why the system would record a value in a particular way. That habit forestalls a great many misreadings.

The Source: The Minnesota Sentencing Guidelines Commission

In 1978 the Minnesota Legislature created the Minnesota Sentencing Guidelines Commission, and in 1980 Minnesota became the first state to sentence felonies under presumptive guidelines developed by such a commission (Frase, 2005; Minnesota Sentencing Guidelines Commission, n.d.). The guidelines derive a presumptive sentence for each case from two facts—the severity of the offense and the offender's criminal-history score—a structure intended to make punishment more uniform and principled (Frase, 2013). To monitor whether courts follow the guidelines, the Commission maintains a record of every person convicted of a felony in adult court and sentenced under them, assembled from probation officers' sentencing worksheets and court data (Minnesota Sentencing Guidelines Commission, 2024). The present dataset is a slice of that monitoring system. For scale, annual felony case volumes have ranged from roughly 5,500 in 1981 to a peak of 18,288 in 2017 (Minnesota Sentencing Guidelines Commission, 2023). This provenance matters for interpretation: the data exists to answer the Commission's question—are sentences following the guidelines?—and was never designed to measure how much crime Minnesota experiences, or why. Keeping that original purpose in view keeps one's conclusions honest.

The Criminal-Justice Funnel

A sentence is the last event in a long sequence. For a felony sentence to exist, a crime must have been committed, noticed, reported, recorded by police, and investigated; an arrest must have been made, charges filed, and guilt established; and a felony-level sentence must have been imposed. Criminologists picture this progression as a funnel: many events enter at the top, and only a fraction reach the bottom. The conception of criminal justice as a connected system, with cases dropping out at each stage, was articulated by the President's Commission on Law Enforcement and Administration of Justice (1967). The dataset sits at the very bottom of that funnel; it contains only cases that survived every prior stage and ended in a felony sentence. It therefore says nothing directly about crimes no one reported, reported crimes that produced no arrest, arrests that produced no charge or conviction, or offenses resolved as misdemeanors or dismissed. Each of these is a stage of the funnel the data never observes.

The Dark Figure of Crime

Criminologists call the crime that never appears in official records the dark figure of crime: occurrences that are treated as crime but registered in no agency's statistics (Biderman & Reiss, 1967). The dark figure is not small. In the National Crime Victimization Survey, only about 42% of violent victimizations were reported to police in 2022, meaning nearly six in ten went unreported (Morgan & Thompson, 2023). Nor is unreported crime a random sample of all crime; it skews toward less serious incidents, smaller losses, and less injury (Skogan, 1977). The crimes that do climb the funnel toward a felony sentence are thus systematically different from those that do not. Before a single row of the dataset was recorded, in other words, most crime had already dropped out of view—and dropped out selectively. No filter applied after the fact can recover it.

Sellin's Dictum: Distance From the Act

In 1931 the criminologist Thorsten Sellin articulated a principle now known as Sellin's dictum: the farther a statistic is collected from the criminal act itself, measured in procedural steps, the less reliable it is as an index of crime (Sellin, 1931). Police reports sit closer to the event than arrests, arrests closer than convictions, and convictions closer than sentences. Sentences lie as far down the procedural chain as it is possible to go. By Sellin's rule, felony sentencing data is therefore an excellent record of one thing—what the system decided to do—and a poor proxy for another—how much crime actually occurred. This single idea reframes the entire enterprise. The roughly 294,000 records constitute a rich, reliable account of Minnesota's sentencing decisions; read as a map of crime itself, they are among the weakest measures available. Almost every error in interpreting the data stems from confusing the two.

How the Nation Measures Crime, and How Minnesota's Data Differs

Because no single source captures all crime, the United States relies on two national systems that sit at different points in the funnel. The FBI's Uniform Crime Reporting program, now transitioning to the more detailed National Incident-Based Reporting System, counts crimes recorded by police (Federal Bureau of Investigation, n.d.). The Bureau of Justice Statistics' National Crime Victimization Survey instead asks a national sample of households about crimes they experienced, including those never reported to police (Bureau of Justice Statistics, n.d.-b). The two disagree by design, because they measure different slices of the funnel. The Minnesota data is neither of these; it is a court and sentencing record that sits below both in the funnel. Consequently, before comparing a finding from this dataset to a national crime statistic, one must ask whether the two even measure the same stage. Usually they do not.

The Unit of Analysis: What One Row Represents

A dataset is only as clear as its unit of analysis—what a single row represents. Here, one row is one sentencing record, and the Commission's counting rules define it precisely: when a person is sentenced in the same county for several offenses within a 30-day window, those offenses are combined into a single record described by the most serious offense (Minnesota Sentencing Guidelines Commission, 2018). Two consequences follow, and both are easy to mistake. First, one row is not one person: an individual sentenced in 2004 and again in 2015 occupies two rows, because the data counts sentencing events rather than individuals—the same distinction federal court statistics draw between a case and a defendant (Bureau of Justice Statistics, n.d.-a). Second, one row is not one crime: secondary offenses within a record are collapsed into the most serious one, and, as the funnel shows, most crimes never produce a record at all. Every total in the dataset should therefore be read not as a count of people or of crimes but as a count of sentencing records. The full dataset comprises roughly 294,000 such records described across more than 100 administrative fields—each field a category the Commission chose to record and, by the same token, a reminder of the many things it did not.

What the Dataset Excludes

Combining the funnel with the counting rules makes plain what these records omit. They exclude unreported crime—the dark figure, which constitutes the majority of offenses (Biderman & Reiss, 1967). They exclude crimes that produced no felony conviction, including cases dropped, dismissed, acquitted, or resolved as misdemeanors. They exclude secondary offenses, which are folded into a record's most-serious charge (Minnesota Sentencing Guidelines Commission, 2018). They exclude most white-collar and regulatory harm, which rarely yields a felony sentence—an offense that by definition may carry a term of at least a year. And they exclude context about

people's lives, since the monitoring system does not maintain data on offenders' social or economic circumstances (Minnesota Sentencing Guidelines Commission, 2024). A final caution belongs here: this is an educational dataset, not a prediction engine. It describes sentences that were imposed; it is not a calculator for the sentence a future case should or will receive. The gap between the guideline-recommended sentence and the sentence actually pronounced is itself a subtle matter, taken up in a later chapter.

Why a Pattern Is Not a Verdict: Description Versus Explanation

A sentencing record is the end product of a long chain of human decisions—victims who report or do not, police who arrest or do not, prosecutors who charge or bargain, and judges who sentence. Every stage filters which cases arrive at the bottom (Klepper et al., 1983). Studying only sentenced cases therefore means studying a selected, non-random sample of everything that happened (Bushway et al., 2007). When two groups exhibit different average sentences, one has found a description of the data, not its cause; correlation, as the adage holds, does not establish causation. A landmark review of sentencing research concluded that even after accounting for the two strongest legal factors—offense severity and criminal history—most of the variation in sentences remains unexplained, and that observed group differences cannot by themselves be read as proof of discrimination (National Research Council, 1983). Descriptive differences are real and worth identifying; explaining why one exists is a separate and harder task. The proper reading of a cross-tabulation showing that offenses in one category receive longer average presumptive sentences than those in another is simply that a descriptive difference exists in the data, with the reasons behind it requiring further, careful analysis.

The Stakes: Why Careful Reading Matters

The insistence on caution has a public rationale: crime statistics are unusually easy to misuse, and the consequences play out in policy and opinion. Despite a long-term decline in recorded violent crime since the early 1990s—the rate has fallen by roughly half—large majorities of Americans have repeatedly told pollsters that crime is worsening (Gramlich, 2024). People reason confidently from numbers they have not examined, precisely the failure that data literacy is meant to prevent (Best, 2001). Three questions guard against it: Where in the funnel does this number sit? What is its unit of analysis? And is a pattern being described or explained?

Summary

The core claims of this chapter can be restated compactly. The Minnesota felony sentencing dataset is administrative data—a by-product of running the state's sentencing system rather than a study designed to measure crime. This is the distinction between found and made data: both involve deliberate thought, but for made data the research question shapes the collection, whereas a monitoring dataset like this one is structured around the needs of the system that produced it (Connelly et al., 2016). The data sits at the bottom of the criminal-justice funnel, containing only cases that survived every earlier stage and ended in a felony sentence (President's Commission on Law Enforcement and Administration of Justice, 1967). Most crime lies in the dark figure and never enters the funnel at all, and it is absent non-randomly (Biderman & Reiss, 1967; Skogan, 1977). By Sellin's dictum, this makes the data an excellent record of decisions and a weak proxy for crime (Sellin, 1931). One row is one sentencing record—not one person and not one crime (Minnesota Sentencing Guidelines Commission, 2018). And a pattern in the data is a description, not an explanation; causes demand separate, careful analysis (National Research Council, 1983). Carried into every subsequent chapter, three questions keep interpretation honest: where in the

funnel a number sits, what its unit of analysis is, and whether a pattern is being described or explained.

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Module 2: Driving the Explorer

Driving the Explorer: Filtering, Summarizing, and Cross-Tabulating the Sentencing Data

This chapter is a working tour of the Minnesota Sentencing Explorer, the interface through which the felony sentencing dataset is examined throughout the course. It assumes no statistical background. Its aim is practical: to establish the five operations used by students, which include filtering, reading descriptive statistics, reverting, clearing, and cross-tabulating. Together, these are sufficient to build almost any view of the data. The dataset comprises roughly 294,000 Minnesota felony sentencing records from 2001 through 2019, and, as the previous chapter established, each row is one sentenced count not one person, and not one crime committed. Three columns recur in what follows: *sentyear*, the year the sentence was imposed; *time*, the presumptive sentence in months; and *moc1*, the first character of the Minnesota Offense Code, which names the top-level offense category.

The Starting Point: The Whole Dataset

With no filters applied, every row is in view. The most important quantity on screen is the count, usually written *N*: the number of rows currently included. Because every operation that follows is measured by its effect on *N*, locating that number is the first habit to build. On the full, unfiltered dataset, *N* is approximately 294,000.

Filtering

A filter keeps only the rows that match a condition and hides the rest. The Explorer offers the standard comparisons: equals (eq), does not equal (ne), greater than and greater-than-or-equal (gt, ge), and less than and less-than-or-equal (lt, le). Two properties matter. First, filtering never deletes data; it narrows what is being looked at, and the underlying records remain untouched. Second, filters chain: each new filter applies only to the rows that survived the previous one. The Explorer

preserves this chain as an ordered filter history, and the list shown on screen is that history.

Applying a single filter—for instance, `sentyear >= 2015`, which keeps only counts sentenced in 2015 or later—adds one entry to the history and reduces N below its full-dataset value.

Reading a Column Summary

For a numeric column such as time, the summary reports several descriptive statistics, of which two deserve immediate attention. The mean is the arithmetic average—every value summed and divided by N . The median is the middle value, with half the rows below it and half above. The two answer slightly different questions and are frequently not the same number. Sentencing data illustrates why: a small number of very long sentences can pull the mean upward while barely moving the median. The gap between them is itself informative—a matter a later chapter takes up—but for now it is enough to locate both in the summary and keep them distinct.

Chaining Filters and the Offense-Code Drill-Down

Because filters stack, adding a second filter applies it only to the rows that survived the first; the history grows by one entry, and N can only stay the same or shrink. The Minnesota Offense Code receives special handling. Rather than entering a full code, one drills down one character at a time. The first character, `moc1`, is the top-level offense category—`moc1 = A`, for example, is assault. A drill-down is a filter like any other and joins the same history. Applying `moc1 == A` on top of `sentyear >= 2015` produces a two-entry history, in the order the filters were applied, and a correspondingly smaller N : the second filter retains only the assault counts among the rows the year filter had already isolated.

Revert and Clear

Storing filters as an ordered history makes undoing precise. Revert removes the most recent entry, stepping back one filter. Clear empties the entire history, returning to the full dataset. Neither

operation loses anything: the data is untouched, and any view can be reconstructed by re-applying the same filters in the same order. Reverting the assault drill-down from the two-filter view, for instance, restores exactly the earlier state in which only the year filter remained, and N returns to precisely the value it held before the drill-down was added. That equality is a useful check that revert does what it claims.

Cross-Tabulation

A cross-tabulation splits the data by two grouping columns at once and reports, in each cell, the N , mean, median, and standard deviation of a third, numeric column. It answers questions of the form: how does presumptive sentence length differ across offense categories, year by year? Reading it is a matter of intersection—choosing a row (an offense category) and a column (a year) identifies the single cell describing only the sentences in that category and that year. A representative table might report the mean of time, with sentyear across the columns and moc1 down the rows, so that the cell where assault meets 2018 gives the mean presumptive sentence for assault counts sentenced in that year.

The idea that demystifies the crosstab is that every cell is a filtered dataset one could have built by hand. The assault-in-2018 cell is exactly the view produced by the filter chain `sentyear == 2018` followed by `moc1 == A`; the mean shown in that cell and the mean in the corresponding column summary are the same number, computed from the same rows. A crosstab, in other words, is a compact array of filtered views, not a new kind of object.

One caution carries over from the previous chapter and belongs here from the outset: a difference between two cells is a description of the data, not an explanation of why that difference exists. Establishing that a gap is real and establishing why it exists are separate tasks, and the latter is the subject of a dedicated later chapter.

Summary: The Five Operations

The five operations, stated compactly: a filter narrows the current view and joins the history; the column summary reports descriptive statistics for any column of that view; revert steps the history back by one; clear empties it; and a cross-tabulation compares a numeric column across two groupings, where every cell is itself just a filtered view. Building a specific view is then a matter of composing these operations in order—clearing the history, filtering sentyear to equal 2018, and drilling moc1 into assault, for example, reproduces the assault-in-2018 cell from first principles. Everything in the chapters that follow is assembled from these same moves.

Module 3: What the Numbers Say

What the Numbers Say: Reading Descriptive Statistics

The operations of the previous chapter produce views of the data; this chapter concerns the numbers those views report. Every statistic considered here is a descriptive statistic—a summary of what is in the current view. Descriptive statistics report patterns; they do not explain why a pattern exists, and that line stays bright throughout. The Explorer computes five such summaries. For categorical columns, which hold labels, it reports value counts and percentages. For numeric columns it reports those plus the mean, median, and standard deviation. The columns that recur below are time, the presumptive sentence in months; moc1, moc2, and moc4, successive characters of the Minnesota Offense Code; sex, the sentenced person's sex as recorded by the court; history, the criminal-history score, where zero denotes no prior history; and tweight, the combined severity weight of additional felony counts at sentencing.

The Mode: The Most Common Category

For a categorical column, the natural summary is the mode—the single most frequent value. Reading it means reading three parts together: the value itself (an offense category), its count (how many rows hold that value), and its percentage (that count as a share of all rows in view). On the full, unfiltered dataset, the mode of moc1 is drug offenses, the largest single block of felony sentencing in Minnesota.

Counts Versus Percentages

A raw count answers *how many*; a percentage answers *how common*. These are different questions, and they can disagree. When one group is far larger than another, the larger group will out-count the smaller on nearly everything—even on characteristics that are actually rarer within it—so whenever groups of different sizes are compared, the percentage is usually the honest

measure. Sentence length by sex illustrates the trap directly. Among women, the "year and a day" sentence—recorded as 12.03 months, since 366 days edges past the one-year felony threshold—accounts for a substantial share of all sentences, even though women are a small minority of sentenced counts. Among men, the same 12.03-month value has a far larger raw count but a far smaller percentage. The pattern is well documented in criminal-justice data: women enter the system far less often, and when they do their sentences tend to run shorter. Reading that pattern correctly requires the percentage, which adjusts for how few women appear in the data at all; the raw count, dominated by the size of the male group, would mislead. Why the pattern exists is a genuinely hard question reserved for a later chapter; the present task is only to read it.

Two Kinds of Middle: Mean and Median

Numeric columns admit two measures of center. The mean sums every value and divides by N . The median sorts the values and takes the middle one, so that half the rows fall below it and half above. A minimal example fixes the difference. For the five numbers 2, 3, 4, 5, and 26, the median is 4—the middle value—while the mean is 8, dragged upward by the single large value. That example is, in miniature, the entire behavior of sentencing data.

Consider first a view in which the two agree. Filtering to assault (`moc1 == A`) and then to terroristic threats (`moc2 == 9`) isolates an offense that carries short expected sentences; the genuinely serious cases tend to be charged as something more severe, so the view holds mostly short sentences with few extreme values and no long tail. There the mean and median of time sit close together—the numeric signature of a roughly symmetric distribution.

Clearing the filters and reading time across all felonies produces the opposite. Most sentences are short—drug offenses, the largest block, typically run about a year—but the same view contains murder, where sentences extend to decades. Those rare, very large values form a long right tail,

and the mean consequently sits well above the median. This is the arithmetic of 2, 3, 4, 5, 26 at full scale: the tail pulls the mean while the median holds its position. As a general reading rule, a mean noticeably above the median indicates a long right tail—many short sentences alongside a few very long ones.

Why the Median Is the Usual "Typical," and When the Mean Still Matters

Returning to the example once more and replacing the 26 with a 6 drops the mean from 8 to 4 while leaving the median unchanged: one value moved, and only one statistic moved with it. The median is, in the statistical sense, robust—it resists extreme values—whereas the mean is not, since every value exerts a pull on it and a single decades-long sentence can shift the mean of an entire year of data. Because the long tail in felony sentencing is not an error to be cleaned away but a permanent feature of the system, researchers and careful journalists habitually report the median sentence as the "typical" one. Encountering a claim that "the average sentence for some offense is 40 months" should therefore prompt an immediate question: mean or median?

The mean is not thereby a mistake to be avoided; it answers a different question. Because the mean multiplied by N yields the total, it is the right instrument whenever total burden is what matters. A corrections department projecting prison capacity cares about the total months to be served, and a budget office estimating annual cost reasons the same way—both are mean questions. A reporter describing what happens to a typical defendant is asking a median question. The competence being built is not "median good, mean bad" but matching the statistic to the question—typical case or total burden—and noticing which one a given claim is quietly relying on.

Spread: The Standard Deviation

Two views can share a mean and still look nothing alike—one huddled tightly around it, the other scattered across a wide range. The standard deviation measures this, capturing, roughly, the

typical distance of a row from the mean. A large standard deviation means the values are widely scattered and the mean says little about any individual case; a small one means the values huddle close and the mean is a good guess for nearly every case. No formula is needed at this stage; like the mean–median gap, the standard deviation is read by comparison.

Both extremes have already appeared. Across the whole dataset, where sentences range from probation length to decades, the standard deviation of time is enormous—comparable in size to the mean itself. Within terroristic threats, where short sentences bunch tightly, it is small. The statistic and the column are identical in the two views; the only thing that differs is how alike the cases are.

[FIGURE PLACEHOLDER 1: distribution of time, full dataset—wide, right-skewed]

[FIGURE PLACEHOLDER 2: distribution of time, terroristic threats only—narrow, compact]

Two readings follow directly. When the standard deviation of time is nearly as large as its mean, the sentences are widely scattered and the mean is a poor guess for any individual case. When it falls close to zero, nearly every sentence sits near the mean and the cases are highly alike.

Where Spread Comes From: Variance as Composition

The deepest idea in this chapter concerns the origin of spread. Reading tweight—the combined severity weight of additional felony counts—on the unfiltered dataset yields a large standard deviation: sentenced counts differ enormously in how much extra felony weight accompanies them. The question is whether that spread reflects erratic sentencing or a view that mixes very different kinds of cases together. Filtering to history == 0, people with no criminal history, answers it: the standard deviation of tweight collapses to roughly a sixteenth of its former value. Nothing about the measurement changed; the composition of the view did. Within a group of similar cases,

the cases are, unsurprisingly, similar. Much of the spread visible in any large dataset is therefore not randomness but mixture. This is why filtering is more than a search tool—every filter poses a question of the form "how much of the pattern survives when I hold this constant?"—and that move is the heart of a later chapter.

The pieces combine into a single reading. A column reporting a mean of 45 months, a median of 21 months, and a standard deviation of 60 months describes a distribution in which most sentences are fairly short, a few are much longer (the mean sitting above the median marks the long right tail), and the values overall are spread very widely (the large standard deviation).

The Final Rule: Check *N* First

One rule governs all of the above: every statistic is only as trustworthy as the *N* beneath it. The point is best made by an extreme case. Filtering to weapons offenses (`moc1 == W`) and then to the fireworks sub-code (`moc4 == 8`) returns every weapons-by-fireworks felony Minnesota sentenced in roughly two decades—exactly four cases, three for possession of illegal fireworks and one for altering serial numbers. The Explorer will still dutifully compute a mean, a median, and a standard deviation of time to two decimal places, but with *N* = 4 those are not statistics so much as four anecdotes in formal dress. Before trusting any summary, in other words, check *N* first.

Summary

The five summaries, in brief: the mode identifies the most common category; percentages rather than counts give the fair comparison when groups differ in size; the median reports the typical case while the mean reports totals; the gap between them indicates the shape of the distribution; and the standard deviation measures spread—spread that often reflects mixture rather than chaos. Beneath all five sits the standing precaution to read *N* before trusting anything the statistics appear to show.

Unit II: The Machine, Minnesota's Sentencing System

Module 4: Classification, Felonies, & The Offense Code

Intro

Felony? Misdemeanor? Criminal justice and law are fields that lean heavily on categorization and classification of human behavior, ultimately messy and at times difficult to cleanly assign.

So, why do we classify crimes?

Classification serves a purpose for those at the entry point of the funnel. Police ... why ()

The legal and procedural background behind classification intertwined with the prosecution system, a charging decision is tied to probable cause to believe that the accused has violated a specific statute (CRS, 2024). These statutes themselves are categorization, critically they divide crimes into felony or misdemeanor. For the purposes of the data you're exploring this is especially useful, all of the cases within are felonies, crimes specifically prescribed the most severe and carrying a sentence of over a year. This data is used to check and balance sentencing guidelines, providing a record of how the guidelines are being . Felony is a useful line in the sand incorporating all kinds of offenses without the overhead of recording and analyzing every speeding ticket or minor theft.

Further along the process judges use classification when making sentencing decisions. ()

Sentencing and even incarceration is not the end of the usefulness of categorizations like these, they are often still of utility to the public and to the government. For example, some states limit social services for those with prior drug felonies. () Furthermore, employers often ask specifically about what kind of felony an applicant committed; to filter out those with a history of violence or larceny (). Classifications make the answer to that question concrete.

Finally, scholars are able to make use of these classifications to generalize their work. ()

Four Families

The conventional criminological partition of offenses is into four “families” *person* (harm directed at a body), *property* (harm to ownership or possession), *drug* (offenses defined by controlled substances), and *public order* (offenses against collective norms or the functioning of institutions).

() Edge cases, such as *theft by force* are features not limitations, categorization is an act of fundamentally human judgement. ()

Let’s explore some examples of the way these families are represented within this dataset using Minnesota Offense Codes. The Minnesota Offense Code or MOC is a system used in Minnesota (where our data is from) to classify crime internally on a system level. We’ll explore one top level code that exemplifies each family to get an idea of how each type of can be classified.

Murder and assault get a lot of coverage in the media (source). However, even collectively they make up a surprisingly small percentage of even just felonies – the most severe crimes.

- Task: look at statistics for moc1, answer questions about how much of the total categories like homicide or assault make up
- Task: now dig into the MOC explorer, find that when moc1=A, for assault a vast amount of those are terroristic threats or verbal assault which isn’t what you might assume the category is primarily consistent of.

Downsides to Classification

As you’ve seen these classifications are important, but they are not without criticism.

Some scholars point to the internal classification of statutes into felony/gross misdemeanor/misdemeanor as having drifted from the ideal of providing a guide to relative

severity. A research team from Pennsylvania conducted an experiment that highlights this criticism. They took crimes and surveyed a representative sample of the public, finding that in many cases crimes the public viewed as equivalent often had categorization significantly different. The difference is not just academic, in one example possessing wiretapping equipment was perceived as being similar to stealing between \$50 and \$200, a maximum guideline sentence of 2 years but in practice the crime is a felony carrying up to seven years in prison. (Paul, et al., 2010).

We also touched on the ways these classifications can be used to limit access for people who fit in those boxes, but while there are very real benefits to employers and agencies the harm it can cause is real as well. ()

References

Congressional Research Service (2024). *Federal Prosecutorial Discretion: A Brief Overview* (LSB11326). Washington, DC: Library of Congress. Link: <https://www.congress.gov/crs-product/LSB11326> A concise legal treatment establishing that "so long as the prosecutor has probable cause to believe that the accused committed an offense defined by statute, the decision whether or not to prosecute, and what charge to bring... generally rests entirely in his discretion." This is the legal backbone for understanding how charging discretion produces classification variability.

Davis, A. J. (1998). *Prosecution and Race: The Power and Privilege of Discretion. Fordham Law Review*, 67(1), 13–67. Link: <https://ir.lawnet.fordham.edu/flr/vol67/iss1/2/> A highly cited law-review analysis arguing the charging decision "almost predetermines the outcome of a criminal case." It documents how discretionary charge selection (including "overcharging" and charge bargaining) shapes which statutory categories are ultimately applied.

Paul H. Robinson, Thomas Gaeta, Matthew Majarian, Megan Schultz & Douglas M. Weck, *The Modern Irrationalities of American Criminal Codes: An Empirical Study of Offense Grading*, 100 J. Crim. L. & Criminology 709–764 (2010) (Vol. 100, Issue 3).

Module 5: The Guidelines Grid

Minnesota as the nation's first presumptive-guidelines state (1980); the grid: ****offense severity × criminal-history score → presumptive sentence****; ``time`` as the grid's output.

Reprise variance-as-composition at full strength: inside one grid cell, the presumptive sentence barely varies — **the grid is visible in the std**. Lab: reconstruct two grid cells by filtering and compare their summaries.

Assumes working severity-level column; ``history`` confirmed. If severity is absent, refit around ``history` × offense family`.*

Module 6: Criminal History — The Second Axis

How the history score accumulates (prior felonies, custody status, decay rules — kept conceptual); the distribution of ``history`` in the data (heavily bottom-weighted); how median ``time`` climbs with ``history`` **descriptively**. First rehearsal of precise language: "sentences are longer at higher history scores" vs. "history causes longer sentences" — here the causal claim is actually **by design** (the grid encodes it), which makes this the safe place to teach the difference between a pattern and a mechanism.

Data anchor: median `time` by `history` level, and the share of counts at `history == 0`.

Module 7: What `time` Measures — Presumptive vs. Pronounced

The gap between the guideline recommendation and what happens: stayed vs. executed sentences,

dispositional and durational ****departures****, plea bargaining's shadow. Lab: audit how stayed / probationary outcomes are coded in `time` (the ZERO-CHECK from Module-3 development) and, if

departure/disposition columns exist, describe departure rates by offense family.

Hard dependency on disposition/departure columns. If absent, this module contracts to a one-week "coding audit + limitations" chapter — which is still genuinely valuable.*

Unit III — Reading Patterns Responsibly

Module 8: Change Over Time

`sentyear` as a variable; volume and median-sentence trends 2001–2019; anchoring trends to policy events — centrally the ****2016 Minnesota drug sentencing reform**** (verify effective dates and which offenses; the dataset holds ~3 post-reform years). The lesson in method: a discontinuity that lines up with a law is **suggestive**, not proof — composition changes too.

Data anchor to hunt: drug-offense median `time` and volume before/after 2016.

Platform prerequisite: this is the module where the absence of line charts hurts most — schedule after visualization ships, or lean on small year-by-year tables.

Module 9: Holding Things Constant

The course's methodological hinge. Comparing groups fairly: composition effects, the filter-as-control move ("how much of the pattern survives when I hold this constant?"), and — if the data yields one — a real ****Simpson's-paradox**** example where a gap reverses inside subgroups. Lab: take a raw gap, add one control filter at a time, narrate what happens to the gap at each step.

Data anchor to hunt: a genuine sign-reversal or major attenuation example; offense-mix differences between two groups are the most likely source.

Module 10: Small Numbers, Big Mistakes

Expansion of check-N-first: rare offenses, single-county single-year cells, the volatility of small-cell means across adjacent years (same filter, consecutive years, wildly different means — nothing changed but noise). The fireworks three return as old friends.

Data anchor: one rare offense whose year-to-year mean swings absurdly.

Module 11: Sex and Sentencing

The gender gap as the **methodological dress rehearsal** for Unit IV: women are rarer in the system and sentenced more shortly — now apply Module 9's toolkit. Does the gap survive holding offense family constant? History score? What remains, and what **can't** this data say about why? The pattern is among the most robust in the sentencing literature, which makes it the right place to practice disparity reasoning with lower interpretive stakes.

Builds directly on the Module-3 base-rate example.

Unit IV — The Big Question

Module 12: Race and Sentencing Disparity

The course's reason for its own caution. ****Disparity vs. discrimination**** as the load-bearing distinction; the naive `race` crosstab, then the Module-9 procedure applied with full care: offense family, severity, history — watching the gap change and being precise about what the residual does and does not establish (unobserved confounders, upstream charging and plea decisions, selection into the dataset). Descriptive data can **document** disparity; it cannot, alone, adjudicate discrimination.

*Prerequisites: Modules 9 and 11 completed; educator teaching-notes feature shipped

(EDUCATOR_PORTAL.md, feature 12); every claim in the narration verified against the data.*

Module 13: What the Data Can't Tell You — and Who Can

The honest ceiling. How researchers go beyond description (conceptual-only tour: regression, matching, the idea of "controlling for" done formally); how to read a sentencing-disparity claim in the news and locate its evidence type; revisiting the funnel — every analysis in this course is conditional on who reached sentencing. Closes the interpretive arc opened in Module 1.

Unit V — Capstone (Weeks 14–15)

Module 14: Designing a Descriptive Investigation

Students pose their own answerable question, pre-specify a filter plan, run it, and check N at every node. The platform's deep feature becomes the pedagogy: ***the filter history is** the methods section** — reproducible by anyone who replays the chain.

*Platform prerequisites: checkpoint wiring (student-built states must be verifiable) and

CSV export.*

Module 15: Communicating Findings

Writing with statistics: choosing mean vs. median and defending the choice; uncertainty language; precise descriptive claims (no causal leakage); a short written report whose appendix is the filter-token chain. Peer review against a rubric drawn from Modules 1, 9, 10, and 12.